SENG 300 - Intro to Software Engineering

Project Milestone I

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Clarification Questions

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1. **Can attendance responses be recorded while an event is still a DRAFT?**

   Assumptions:
   - Response recording is only offered for SCHEDULED events, ACCEPTED needs a fixed time to check availability/conflicts, and a draft has no time.
   - An invitee added to a draft starts as PENDING
   - P1 only stores the PENDING default.

   Unresolved Risks:
   - Cancelled events keep their responses but it is unclear whether those responses can still be edited.
   - If the 3 sample events include scheduled ones with ACCEPTED responses, they must obey the conflict rules since otherwise the app would reject its own sample data.

   Answer: Responses are only recorded after the event is scheduled so draft events keep every response pending.

2. **After an event is scheduled, which fields count as the fixed "planning fields" ?**

   Assumptions:
   - Planning fields are title, invitees, date window, required features, duration, minimum attendance, room, start/end time. Responses and status change.

   Unresolved Risks:
   - P3 adds controlled scheduling which changes the room or time on the same event, keeps old responses as read-only and resets responses to PENDING if date/time changes. Cancelling/recreating event gives it a new ID and loses that history, so it won't meet P3.

   Answer: All fields are fixed except responses. To change a plan, cancel and create a new event.

3. **Is student search case-insensitive? And does it match anywhere in the name/email?**

   Assumptions:
   - Search text is trimmed first, and whitespace-only search counts as empty.
   - A student matches if the text appears anywhere in the name/detail.
   - Results come in a fixed order so repeated searches match.

   Unresolved Risks:
   - Case-insensitive matching in Java can depend on locale. Use a fixed locale and add a unit test with mixed-case input.

   Answer: Yes. An empty search lists everyone.

4. **Who is responsible for saving and loading application data?**
   Assumptions:
   - Nothing is saved automatically. Changes stay in memory until the organizer chooses Save.
   - Save writes students, availability and draft events in one operation.
   - Load replaces what is in memory.

   Unresolved Risks:
   - With a manual save, changes not saved before closing are lost. Acceptance criteria, tests, and demo must include the save step.
   - Unclear whether startup loads file automatically or only when the organizer chooses Load.
   - A corrupt file shouldn't wipe the data in memory or get overwritten. This needs a test.

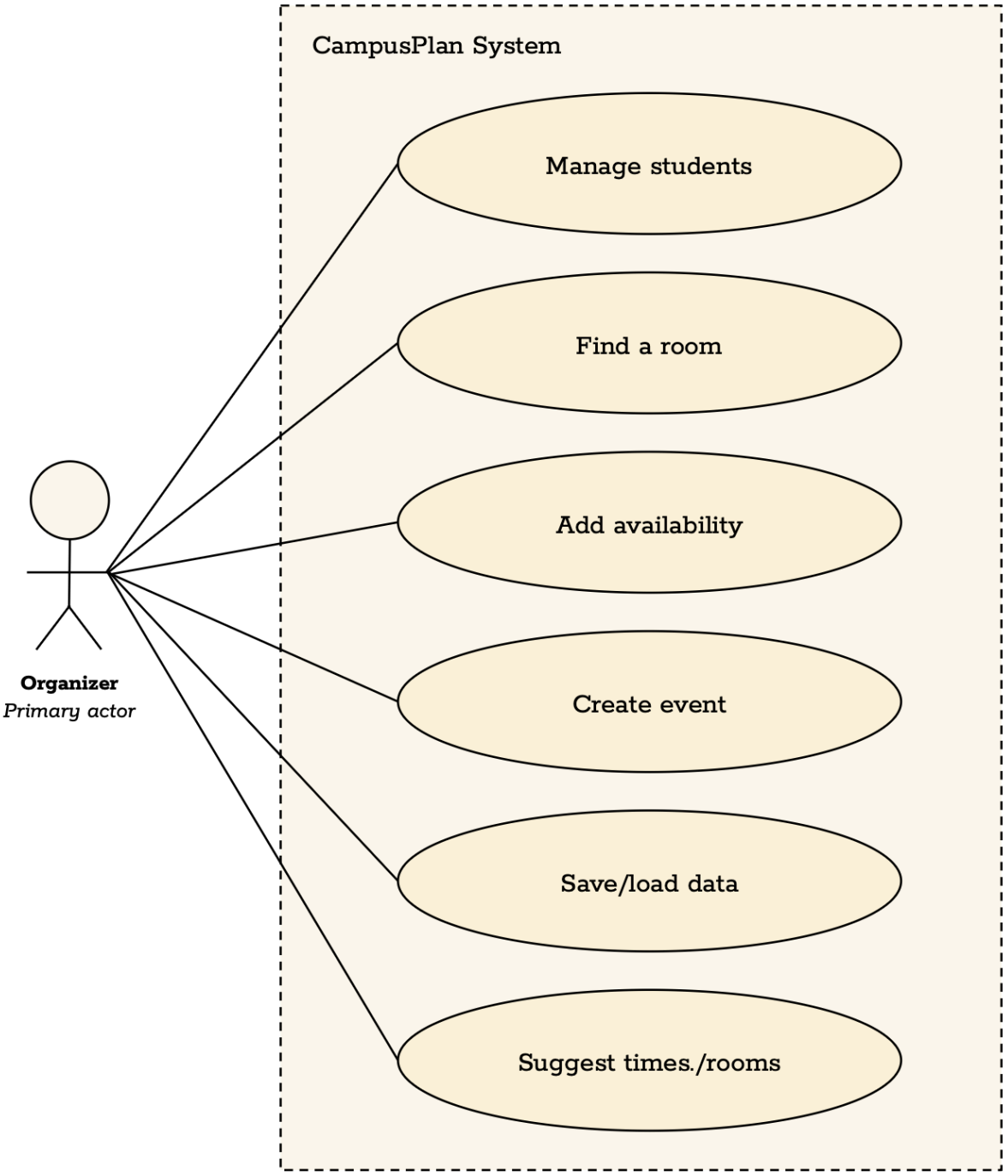
   Answer: The organizer manually saves and loads application data. The system ensures that saved data persists after restarting.

5. **Can the organizer remove previously entered availability intervals?**
   Assumptions:
   - Removal covers a whole interval or part of one, and removing part can split an interval.
   - Only ACCEPTED responses on SCHEDULED events block a removal. PENDING, DECLINED, drafts, and cancelled events don't.
   - Blocked removal changes nothing and names the affected events.

   Unresolved Risks:
   - No responses can be recorded in P1. If P1 removes intervals, P2 has to add the check later.
   - If the sample data includes ACCEPTED responses, add a test that shows a conflicting removal is rejected.

   Answer: Yes. The organizer can remove availability intervals provided the removal does not conflict with any accepted event attendance.

Use Cases Diagram



Use Cases Details:

UC 1: Manage students
The organizer adds, edits, views, and searches for students while the system checks each name/email and ensures only valid and non-duplicate entries exist.

Alternatives:

- A1: The organizer changes the name/email and the system correctly validates the changed fields. A student's current email does not count as a duplicate of itself.
- A2: The organizer lists all the students and can see the details if they open a student.
- A3: The organizer enters a name/email and the system shows every match. The system shows nothing if otherwise.
- A4: Entries - The organizer does not save any entry before confirming.

Errors:

- E1: Name is longer than 100 characters or blank after trimming.
- E2: Email does not have an @ symbol, no text on either side of the @, or contains whitespace.
- E3: The organizer asks to edit or view a student that does not exist.
- E4: Email duplicates another student's email after trimming/lowercasting.

UC 2: Find a room
The organizer views the room catalog and can filter it by capacity and required features to view rooms that fit their event.

Alternatives**:**

- A1: Only a feature filter is used. Capacity is empty.
- A2: Only a capacity feature is used. The feature is empty.
- A3: The system shows the full catalogue again when the organizer clears the filters.
- A4: The system shows no room matches and lets the organizer change the current filters.
- A5: Catalog is empty because storage is missing and seed data was not loaded. The system tells the organizer and points to the seed-data load in UC 5.

Errors:

- E1: Capacity is not a valid number, has decimals or user entered text.
- E2: Capacity is < 1 or >100. Must be 1-100.
- E3: A feature name is neither of WHITEBOARD, ACCESSIBLE, or PROJECTOR.

UC 3: Add availability
The organizer enters and updates a student's availability, with missing availability indicating no availability.

Alternatives:

- A1: The organizer edits an existing availability interval and the system updates the students availability.
- A2: The organizer enters an interval that overlaps or touches an existing interval; the organizer merges the intervals into one continuous interval.
- A3: The organizer enters multiple valid intervals on the same date; the system stores all intervals.

Errors:

- E1: Availability start does not precede end.
- E2: Availability interval falls outside opening hours (8:00-22:00).
- E3: Room availability is not within 30-minute boundaries.
- E4: Availability interval spans more than one date.
- E5: The organizer enters availability for a student that does not exist.
- E6: Updating availability would invalidate an existing ACCEPTED attendance; the system rejects the change and identifies the affected event(s).

UC 4: Create event
The organizer creates a draft event by entering a title, invited students, date window, duration, required room features, and minimum attendance.

Alternatives:

- A1: The organizer edits an existing DRAFT event and the system updates the event details.
- A2: The organizer creates an event without selecting any room features. Required room features may be empty.
- A3: The organizer views the details of an existing event after creation.
- A4: The organizer cancels event creation before confirming. No event is saved.

Errors:

- E1: The title is empty after trimming or exceeds 100 characters.
- E2: Fewer than 2 students or more than 20 students are invited.
- E3: The invitee list contains duplicate students.
- E4: The date window is less than 1 day or greater than 7 consecutive days.
- E5: The duration is less than 30 minutes, greater than 180 minutes, or not a 30-minute increment.
- E6: Minimum attendance is less than 2 or greater than the number of invited students.
- E7: One or more selected students do not exist.

UC 5: Save/load data
The organizer saves and reloads application data, including students, rooms, availability, and draft events. The system stores the data and preserves record IDs so that information remains available after restarting the application.

Alternatives:

- A1: No saved data file exists. The system starts with empty data and offers the organizer the option to load the provided seed dataset.
- A2: No saved data file exists, and the organizer chooses not to load the seed dataset. The system remains empty and allows the organizer to enter new records.
- A3: The organizer loads the seed dataset into an empty application. The system initializes the sample students, rooms, availability, and events.

Errors:

- E1: The saved JSON file is corrupt or invalid. The system displays a clear error and does not overwrite the existing file.
- E2: The system fails to save data due to a storage error. The system reports the failure and does not discard the previous valid file.
- E3: The organizer attempts to load seed data into an application that is not empty. The system rejects the operation to prevent overwriting or duplicating existing data.

UC 6: Suggest times/rooms
The organizer requests scheduling suggestions for an event. The system generates and ranks suitable combinations of times and rooms based on room requirements, student availability, scheduling conflicts, and minimum attendance. The system displays up to five suggestions, including eligible and ineligible students and reasons for ineligibility.

Alternatives:

- A1: Fewer than five valid suggestions are available. The system displays all qualifying suggestions.
- A2: No valid suggestions are found. The system informs the organizer and displays the number of candidates rejected due to room constraints and insufficient eligible attendance.
- A3: Multiple suggestions have the same number of eligible students. The system ranks them by earliest date/time, fewest unused room seats, and room ID.
- A4: Some invited students are unavailable or have conflicting accepted attendance. The system excludes those students from the eligible count, provides reasons, and still displays suggestions that meet minimum attendance.

Errors:

- E1: The organizer requests suggestions for an event that does not exist. The system rejects the request.
- E2: A room does not meet the required capacity or features, or conflicts with another scheduled event. The system excludes that room/time candidate.
- E3: The number of eligible students is below the event's minimum attendance. The system excludes the candidate from its suggestions.